
LOAN DEFAULT PREDICTION

Executive Business Summary

Prepared by: Nainil Shah

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Model Accuracy: 81.3% | ROC-AUC: 0.867



PROBLEM STATEMENT

Financial institutions face 25-30% loan default rates, resulting in significant revenue loss. This ML model predicts default risk before loan approval to minimize losses and optimize lending decisions.



KEY FINDINGS

1. CREDIT HISTORY = STRONGEST PREDICTOR (10x Impact)

- Customers WITH credit history: 79.6% approval rate
- Customers WITHOUT credit history: 7.9% approval rate
- Impact: 10 times difference in approval likelihood

2. HIGH-RISK SEGMENT IDENTIFIED

- 156 customers (25.4% of total) classified as High/Very High Risk
- These customers represent potential loss of Rs 68.5 Lakhs
- Current approval rate for high-risk: 35% (too high!)

3. SELF-EMPLOYED = HIGHER RISK

- Self-employed: 64.3% approval vs Salaried: 70.2% approval
- 6% higher rejection rate indicates income stability concerns
- Requires stricter verification process

4. GEOGRAPHIC RISK VARIATION

- Rural areas: 64.8% approval (lowest)
- Semiurban areas: 70.8% approval (highest)
- 6% difference suggests location-based risk factors



BUSINESS RECOMMENDATIONS

IMMEDIATE ACTIONS (0-30 days):

✓ 1. MANDATORY CREDIT HISTORY CHECK

The bank should reject those customers who don't have a credit history because their approval rate is only 8% and 92% defaults occur.

Expected Impact: 15-20% reduction in defaults

✓ 2. DEPLOY ML MODEL IN LOAN APPROVAL PROCESS

Deploy the model in production and automatically score every application. Put high-risk customers (score < 40%) into auto-reject or manual review.

Expected Impact: Save 5-7 Lakhs annually

✓ 3. STRICTER VERIFICATION FOR SELF-EMPLOYED

Through Self-employed applicants :

- 12 months bank statements mandatory
- Business proof verification
- Higher down payment (25% vs 15%)

Expected Impact: 10% reduction in self-employed defaults

MEDIUM-TERM ACTIONS (30-90 days):

✅ 4. RISK-BASED PRICING

- Rural area loans: +0.5% interest rate premium
- No credit history: +1% interest rate OR additional collateral
- High loan-to-income ratio (>5%): Require guarantor

✅ 5. LOAN AMOUNT CAPS

Maximum loan = 3x annual income (currently some loans are 5x income)

FINANCIAL IMPACT & SAVINGS

CURRENT STATE:

- Total Applications Analyzed: 614
- Current Approval Rate: 68.7%
- High-Risk Customers Approved: ~70 (estimated)
- Average Loan Amount: Rs 1.46 Lakhs
- Default Rate (industry avg): 15%
- Loss per Default: Rs 43,800 (30% of loan amount)

POTENTIAL LOSS (Without ML Model):

- High-risk approvals: 70 customers
- Expected defaults (50%): 35 customers
- Loss per default: Rs 43,800
- TOTAL POTENTIAL LOSS: Rs 15.3 Lakhs per year

SAVINGS WITH ML MODEL:

- Model correctly identifies 81% high-risk customers
- Prevented high-risk approvals: 57 customers (81% of 70)

- Prevented defaults: 28 customers (50% of 57)
- Loss avoided: $28 \times \text{Rs } 43,800 = \text{Rs } 12.26 \text{ Lakhs}$

 ESTIMATED ANNUAL SAVINGS: Rs 5-7 Lakhs

(Conservative estimate accounting for false positives)

ROI: 300%+ (Model development cost vs. savings)

MODEL PERFORMANCE SUMMARY

- Algorithm: Random Forest Classifier
- Accuracy: 81.3%
- Precision: 84.2% (84% of predicted approvals are correct)
- Recall: 90.5% (catches 90% of actual good customers)
- ROC-AUC: 0.867 (excellent discrimination)

Top 3 Predictive Features:

1. Credit History (28.5% importance)
2. Total Income (15.6% importance)
3. Loan Amount (14.2% importance)

PROJECT LINKS

Live Demo: <https://loan-predictor-nainil.streamlit.app>

GitHub: <https://github.com/Nainilshah04/loan-default-prediction>

Built by: Nainil Shah | nainilshah04@gmail.com
